

Total No. of Questions : 10]

SEAT No. :

P2978

[5154]-532

[Total No. of Pages : 3

**B.E. (Mechanical Engineering)
CAD/CAM and Automation
(2012 Course) (End Semester) (402042)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer five questions.*
- 2) *Figures to the right indicate full marks.*
- 3) *Use of electronic calculator allowed.*
- 4) *Assume suitable data.*

Q1) A line is drawn with vertices A(4,4) B(15,10) has undergone following transformation in sequence. **[10]**

- a) Rotation through 45° anticlockwise
- b) Scaling by 1.5 units.
- c) Reflection about x axis.

Find concatenated matrix and new co-ordinates of triangle.

OR

Q2) a) Write equation of line having $P_1 [3,5,8]$ and $P_2 [6,4,3]$ and find tangent vectors and points on line at $\mu = 0.25, 0.5, 0.75$. **[5]**

b) Write short notes on Hermite cubic spline curve with neat sketch. **[5]**

P.T.O.

- Q3) a) Explain feature based modelling. [4]
 b) Find displacement and reaction force for cluster of springs as shown in fig.1 [6]

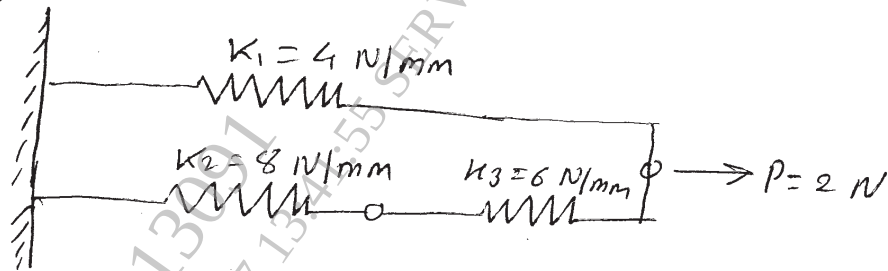


Figure 1

OR

- Q4) Truss element is as shown in figure 2. Determine, deflection, stresses and reaction force in element. [10]

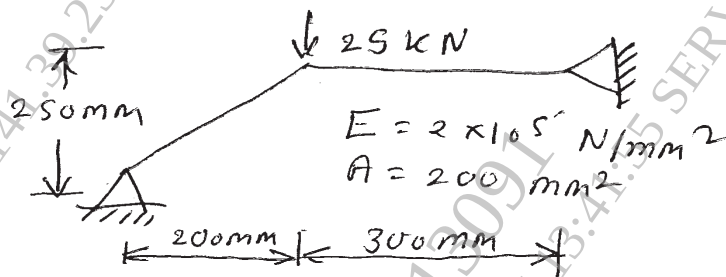
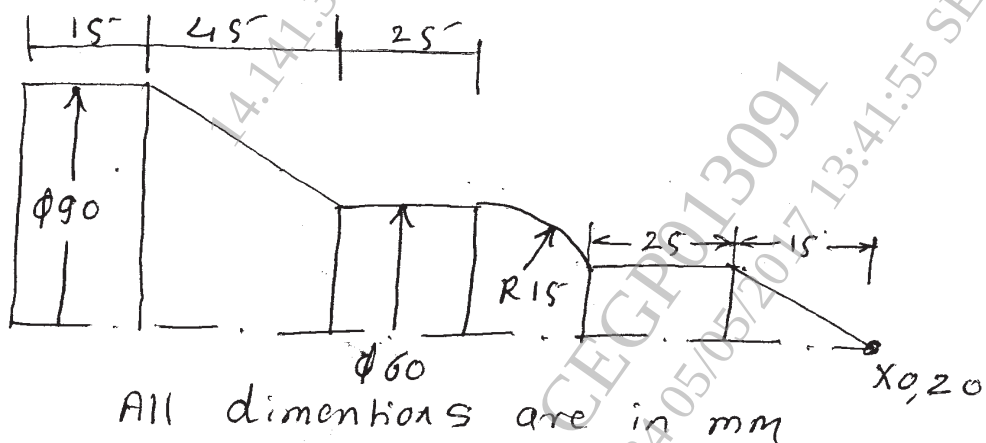


Figure 2

- Q5) a) Explain canned cycle for parting and threading for turned components. [6]
 b) Write CNC port program for roughing and finishing using canned cycle for turned components as shown in figure 3. Assume suitable data [12]

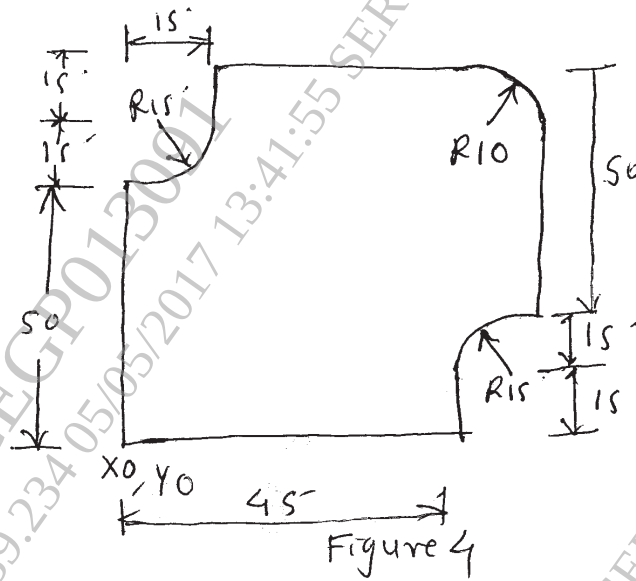


All dimensions are in mm

Figure 3

OR

- Q6)** a) Explain concept of sub programming. [6]
 b) Write CNC program for the component shown in figure 4. [12]



- Q7)** a) Explain basic steps in R.P. process. [6]
 b) Explain fused deposit manufacturing R.P. process. [10]

OR

- Q8)** a) Explain laminated object manufacturing (LOM) modelling method of R.P. [12]
 b) Explain of R.P. Aerospace Industry. [4]

- Q9)** a) Explain various joints used in robots with neat sketch. [8]
 b) Explain the vacuum gripper with neat sketch and its applications. [8]

OR

- Q10)** a) Explain Automation strategies. [8]
 b) Explain Group Technology layout and advantages over process layout. [8]

x

x

x